TEXORA · LECTURE SHEET

Cogeneration Plant

A dyeing mill needs electricity and process steam at the same time, all day. Cogeneration — combined heat and power — produces both from one fuel stream and uses heat that a power-only plant throws away. This sheet covers topping and bottoming cycles, working principles, a fully worked efficiency comparison, benefits and applications.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 4 · LEVEL 4 TERM 1

1. Definition

Cogeneration (combined heat and power, CHP) is the simultaneous production of two useful energy forms — typically **electricity (or shaft power) and useful heat (process steam or hot water)** — from a single fuel input in one integrated plant. In a conventional power plant the condenser rejects roughly two-thirds of the fuel heat (Topic 2, 33.2% example). In cogeneration that low-grade heat is not rejected: it is delivered to a process that needs it, so overall fuel utilisation commonly reaches 70–85% in well-matched plants (illustrative textbook range).

$$\text{Overall (CHP) efficiency} = (\text{electricity out} + \text{useful heat out}) \div \text{fuel energy in} \times 100\%$$

The plant is economic only when the heat is genuinely used as it is produced — the **heat load and power load must coincide in time and be reasonably balanced**. A textile wet-processing mill is an almost ideal host: dyeing and finishing run long hours and consume large, steady quantities of low-pressure steam while the same site draws megawatts of electricity.

2. Topping cycle

In a **topping cycle**, fuel energy is used *first* at high temperature to generate power, and the lower-temperature heat leaving the power equipment is then used by the process. Power is the primary (top) product; process heat is recovered from its exhaust.

Forms: (a) **Back-pressure steam turbine:** boiler steam expands in the turbine only down to the process pressure (not to condenser vacuum); the exhaust steam goes straight to the process. Every kg of steam gives both work and then process heat. (b) **Gas turbine / gas engine with heat recovery:** exhaust gases pass through a heat-recovery boiler raising process steam; engine jacket water and oil coolers can supply hot water. (c) **Extraction-condensing turbine:** part of the steam is extracted at

process pressure, the rest condenses — more flexible when heat and power demands vary independently.

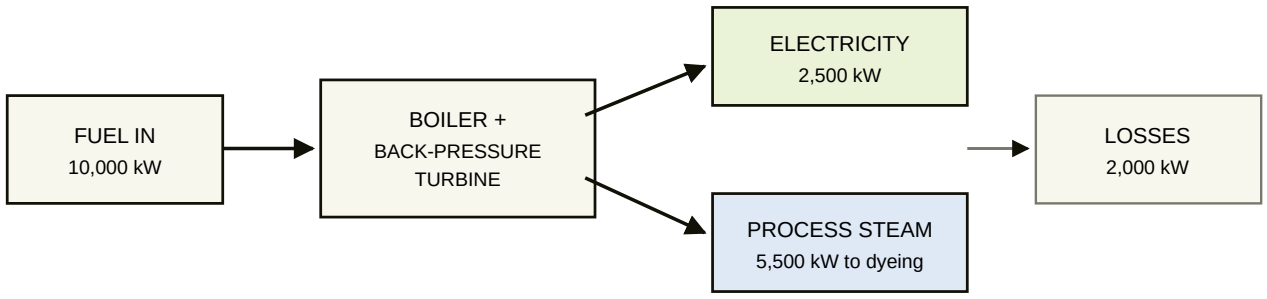


Figure 1 — Topping-cycle block diagram (back-pressure steam turbine). Fuel 10,000 kW → electricity 2,500 kW + process heat 5,500 kW + losses 2,000 kW.

3. Bottoming cycle

In a **bottoming cycle** the order is reversed: fuel energy is used *first* at high temperature by an industrial process (a furnace, kiln or reactor), and the hot waste stream leaving that process is used to raise steam and generate power. Power is recovered at the bottom of the temperature cascade. Bottoming cycles suit industries with very hot exhaust streams — cement kilns, steel and glass furnaces, large chemical reactors. In textiles, true bottoming is uncommon because process temperatures are modest; small waste-heat boilers on engine exhausts or stenters are heat recovery, and when paired with generation they form part of a topping arrangement in most mill layouts.

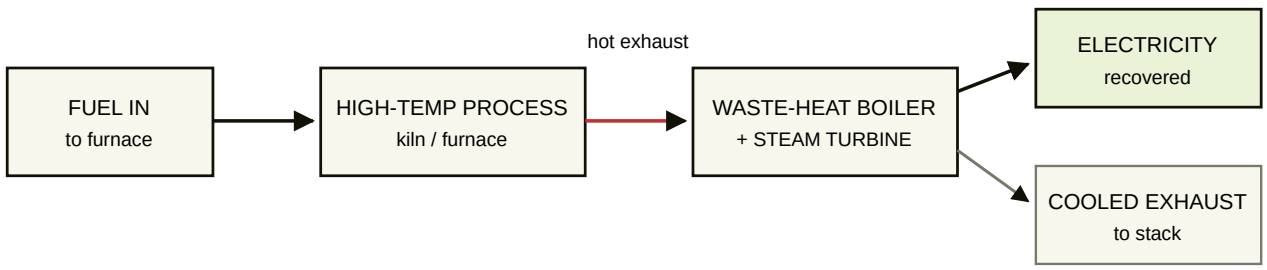


Figure 2 — Bottoming-cycle block diagram. Fuel serves the high-temperature process first; the waste-heat boiler and turbine then generate power from the exhaust.

4. Worked example — overall efficiency and the saving over separate generation

WORKED EXAMPLE — BACK-PRESSURE COGENERATION VS SEPARATE SUPPLY (ALL VALUES ILLUSTRATIVE, INTERNALLY CONSISTENT)

Cogeneration plant: fuel input = 10,000 kW (thermal). Generator output = 2,500 kW. Exhaust steam delivers useful process heat = 5,500 kW. Stack, radiation and other losses = 2,000 kW. Check: 2500 + 5500 + 2000 = 10,000 kW ✓.

Overall efficiency = (2500 + 5500) / 10,000 = 8000/10,000 = **80.0%**.

Separate generation of the same outputs: electricity from a power-only plant at 35% efficiency needs 2500/0.35 = 7,143 kW of fuel. Process steam from a boiler at 85% efficiency (Topic 3 range) needs 5500/0.85 = 6,471 kW of fuel. Total separate fuel = 7143 + 6471 = 13,614 kW.

Fuel saving = (13,614 − 10,000)/13,614 ≈ **26.5%** less fuel for identical electricity and steam — and the separate route's combined utilisation is only 8000/13,614 ≈ 58.8%, against cogeneration's 80.0%.

Working principle, step by step (steam topping plant): (1) The boiler raises high-pressure superheated steam. (2) Steam expands in a back-pressure turbine to the process pressure, driving the generator. (3) Exhaust steam at process pressure flows to dyeing, drying and heating. (4) Condensate returns to the feed tank; make-up is treated. (5) If process steam demand falls, a pressure-reducing and desuperheating station or a dump condenser trims the balance — a design detail that decides real-world savings.

5. Benefits

- **Fuel and cost saving:** one fuel stream does two jobs (26.5% in the worked example).
- **Lower emissions per unit of useful energy**, because less fuel is burnt for the same output.
- **Supply security:** captive generation keeps critical loads running during grid disturbances; steam and power are under the mill's own control.
- **Reduced transmission losses** — power is generated where it is used.
- **Smaller separate boiler duty** and better boiler loading, which itself improves efficiency.

Limits: high capital cost; savings collapse if the heat load disappears (holidays, low production); gas supply and tariff conditions decide payback; skilled operation is essential.

6. Applications

- **Textile wet processing:** dyeing and finishing mills — gas-engine or steam-topping CHP supplying dye-house steam and mill electricity; exhaust-heat boilers on stenters and engines.
- **Paper, sugar, chemicals, food processing:** all combine steady process steam with large power demand.

- **District heating and institutions:** hospitals and campuses use CHP heat for heating, laundry and sterilisation.
- **Bottoming applications:** cement, steel and glass plants recovering kiln/furnace exhaust heat for power.

Glossary

Cogeneration / CHP — simultaneous production of power and useful heat from one fuel input. **Topping cycle** — power generated first; its exhaust heat serves the process. **Bottoming cycle** — process uses heat first; power is recovered from its waste heat. **Back-pressure turbine** — turbine exhausting at process pressure instead of condenser vacuum. **Extraction turbine** — turbine with steam tapped off at an intermediate pressure. **HRSG / waste-heat boiler** — boiler raising steam from a hot exhaust stream. **Overall efficiency** — (power + useful heat) ÷ fuel input.

Key points

• Cogeneration uses the heat a condensing plant rejects; overall efficiency = (power + useful heat)/fuel.
• Worked example: (2500 + 5500)/10,000 = 80.0%, versus 58.8% utilisation for separate supply — a 26.5% fuel saving.
• Topping = power first (textile mills). Bottoming = high-temperature process first, power from waste heat (cement, steel, glass).
• CHP pays only when heat and power loads coincide — the dye house is the textbook host.

Practice

1. A CHP plant takes 8,000 kW of fuel, generates 2,000 kW of electricity and supplies 4,400 kW of process heat. Find its overall efficiency and its total losses.
2. For the plant in Q1, calculate the fuel needed to supply the same outputs separately (power at 33%, steam at 84%) and the percentage fuel saving.
3. A cement kiln exhaust is proposed as a power source. State whether this is topping or bottoming, and why.
4. Give two reasons a cogeneration plant can disappoint in its first year in a real mill.

Answer key

1. Overall η = (2000 + 4400)/8000 = **80.0%**. Losses = 8000 − 6400 = **1,600 kW**.
2. Separate fuel = 2000/0.33 + 4400/0.84 = 6061 + 5238 = 11,299 kW. Saving = (11,299 − 8000)/11,299 ≈ **29.2%**.

3. **Bottoming** — the high-temperature process (kiln) uses the fuel first; power is generated afterwards from its waste heat.
4. Any two of: heat and power loads do not coincide (heat dumped at low production); steam/power balance mismatched to the turbine choice; gas price or availability changed; poor condensate return or maintenance eroding the design efficiency.
